

Editors-in-Chief,
Journal of Functional Analysis
Prof. Alexandru D. Ionescu, Prof. Natasa Sesum, and Prof. Stefaan Vaes

Dear Editors-in-Chief of the Journal of Functional Analysis,

I am writing to submit the manuscript “Singular Gaussian Conditioning via the Inverse-Free Schur Complement” as an original research article for JFA.

In finite dimensions, Gaussian conditioning is represented by the Schur form

$$A - BD^{-1}B^*.$$

In Hilbert-space Gaussian models with trace-class covariance, this form is not a valid ambient operator formula: the complementary covariance block is compact and has no bounded inverse on the full state space. This paper shows that the correct replacement is the Anderson–Trapp shorted operator, understood as an inverse-free Schur construction on the Cameron–Martin range.

Our results provide:

1. a rigorous shorting reformulation of the block-triangular OU regularization and the rank-one conditioning setup;
2. the canonical scaling law $\Sigma_{\lambda,kk} = \varepsilon_0/(2\lambda^2)$ and residual estimate

$$S_{k,\lambda} = \frac{\varepsilon_0}{2\lambda^2} - R_\lambda, \quad R_\lambda = O(\lambda^{-6}),$$

which separates the singular and finite parts;

3. a Galerkin consistency theory via nested Rayleigh–Ritz truncations, avoiding direct matrix inversion;
4. Gaussian finite-part asymptotics for rank-one disintegration,

$$\mathfrak{J}_\lambda(\mu_{\text{obs}}; \mu_\lambda) = \frac{b_k^2}{\varepsilon_0} + O(\lambda^{-4}),$$

obtained from disintegration and Fredholm determinant/trace cancellations.

We also include a concrete 1D elliptic Gaussian field example to demonstrate the framework in a standard trace-class/SPDE setting.

The manuscript is closely aligned with JFA’s scope through its focus on positive block operators, shorted operators, Cameron–Martin geometry, trace-class covariance, and Hilbert-space variational methods; the probabilistic application is a major consequence of the operator-theoretic core.

Preferred handling editors:

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Thank you for your consideration.

Sincerely,

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